Requirements Gathering Document

Project: Smart Clinic — an AI assistant that guides patients before they visit the doctor

Version: 1.0 (Baseline) **Date:** 30 October 2026 **Prepared by:** Smart Clinic team (6 members)

This document records what the project must do, how it must perform, and what limits apply, before development begins. Once signed off in Section 6, it becomes the baseline. Changes after that follow the change-control process in Section 7.

1. Identify Stakeholders

Stakeholder	Role	Influence	Interest	Main need
Project Owner (Instructor)	Approves scope, deliverables, and final result	High	High	A working, well-documented product delivered on time that matches the approved scope.
Patients (primary users)	Use the website with respiratory symptoms	Medium	High	Simple language, clear next step, and the right specialty without confusion.
Doctors and clinics	Receive the patient's summary	Medium	Medium	A short, accurate summary that saves time in the first examination.
Development team	Design, build, test, deploy	High	High	Clear scope, available data, and enough time and tools.
Dataset providers (Kaggle, Open-i)	Supply public data	Low	Low	Respect of dataset licences and proper citation.
Cloud provider (Azure)	Hosts models and website	Low	Low	Use within free or student limits.

Influence / interest map

- **Manage closely (high influence, high interest):** Project Owner, Development team.
- **Keep informed (medium influence, high or medium interest):** Patients, Doctors and clinics.
- **Monitor (low influence, low interest):** Dataset providers, Cloud provider.

Potential conflict spotted early: patients want a clear answer, while the Project Owner and doctors need the output to stay a preliminary guide. This is handled by requirement FR-12 (clear wording and disclaimer on every page).

2. Plan the Approach

Objectives

- Help patients with chest and respiratory symptoms understand their situation and reach the right specialty faster.
- Combine symptoms, a chest X-ray image, and an X-ray report into one simple summary page.

High-level scope

In scope	Out of scope
Chest and respiratory conditions	Other medical specialties

In scope	Out of scope
Symptom input in free text (English and Arabic phrases)	Voice input and image types other than chest X-ray
Chest X-ray: normal vs. pneumonia opinion	Detecting other X-ray findings (for example, tumours, fractures)
Report input as text or file; normal/abnormal result and term explanation	Writing reports or giving a final diagnosis
One downloadable summary page with suggested specialty	User accounts, medical history, or booking appointments
Website published online	Mobile application

Timeline and resources

- **Timeline:** 4 weeks (Week 1 setup and baselines, Week 2 model development, Week 3 best models and website, Week 4 integration, deployment, and final delivery).
- **Team:** 6 team members with defined roles (Data, Symptoms ML, CNN, Transfer Learning, NLP, Website and Cloud).
- **Resources:** free public datasets from Kaggle; Google Colab / Kaggle Notebooks (free GPU); GitHub; MLflow; Azure.

Elicitation techniques selected

- **Structured interview** with the Project Owner to confirm scope and constraints.
- **Short survey / informal interviews** with a small group of potential patients and with a doctor or medical student, if available, to check wording and usefulness.
- **Walkthrough** of the page flow (wireframes) before development.
- **Review** of existing symptom-checker websites.

3. Conduct Elicitation

The questions below focus on needs and underlying problems, not on tools. The answers are the positions agreed for the baseline.

A. Business need and users

Question	Agreed answer
What problem must the product solve?	Patients with respiratory symptoms are confused about what the condition might be and which specialty to visit, causing worry, delays, and wrong visits.
Who will use it?	Patients with respiratory symptoms (primary). Doctors and clinics receive the summary the patient downloads (secondary).
What does success look like?	A patient completes the flow and receives one clear summary with possible diseases, X-ray opinion, report explanation, and a suggested specialty.
Which language should patients use?	Symptoms can be written in English or Arabic phrases (including common Egyptian expressions). Output is in simple language.

B. Inputs, outputs, and behaviour

Question	Agreed answer
What does the patient provide?	(1) Symptoms in free text, (2) a chest X-ray image, (3) the X-ray report as typed or pasted text, or as an uploaded file.
What does the patient receive?	A summary page: extracted symptoms, closest possible diseases with probabilities, X-ray opinion with a confidence value, report result (normal or needs follow-up) with key terms explained, suggested specialty, and a medical disclaimer.
Is the output a diagnosis?	No. It is a preliminary guide. The wording must say it is not a final diagnosis and not a substitute for a doctor.
Must all three inputs be provided?	The full summary uses all three. If one is missing, the page still shows the available results and states which input is missing.
Can the patient keep the result?	Yes, the summary can be downloaded to show to the doctor.

C. Data and models

Question	Agreed answer
Which data is used?	Free public datasets: Disease Prediction Using Machine Learning and Disease Symptom Prediction (symptoms), Chest X-Ray Images (Pneumonia) (X-rays), and IU X-Ray / Open-i (reports).
Do we collect or store patient data?	No. Inputs are processed for the current session only and are not stored permanently.
What quality is expected?	Target accuracy on a held-out test set: symptoms and X-ray at about 90% or higher, report classification at about 85% or higher. These are targets, reported honestly with Precision, Recall, and F1, not guarantees.
What if a better dataset is found?	It may replace a dataset through the change-control process (Section 7).

D. Safety, performance, and deployment

Question	Agreed answer
What must always be shown?	The medical disclaimer on every page.
What if symptoms look urgent (for example, severe shortness of breath or chest pain)?	The page shows a clear warning to seek immediate medical care.
How fast should it respond?	The summary should appear within about 10 seconds after the inputs are submitted.
Where will it run?	Published online on Azure, accessible from a web browser on desktop and mobile.
Are there cost limits?	Free or student-tier resources only.

4. Document Observations and Findings

Functional requirements (what it must do)

ID	Requirement	Component
FR-01	The user can enter symptoms as free text.	Website / Mind 1
FR-02	The system extracts symptoms from the text, including Arabic and Egyptian phrases, using a dictionary and rules.	Arabic layer
FR-03	The system predicts the closest three possible diseases with probabilities.	Mind 1 (ML)
FR-04	The system maps the predicted diseases to a suggested specialty.	Mind 1 / Integration
FR-05	The user can upload a chest X-ray image (JPG or PNG).	Website
FR-06	The system classifies the image as normal or pneumonia and shows a confidence value.	Mind 2 (DL/CV)
FR-07	The system shows a heatmap of the regions that influenced the X-ray result.	Mind 2
FR-08	The user can type, paste, or upload the X-ray report text.	Website
FR-09	The system classifies the report as normal or needs follow-up.	Mind 3 (NLP)
FR-10	The system extracts important medical terms from the report and explains them in simple words.	Mind 3
FR-11	The system shows one final summary page and lets the user download it.	Website / Integration
FR-12	Every page shows the medical disclaimer, and results are worded as guidance, not diagnosis.	Website
FR-13	The system shows a warning to seek immediate care when urgent symptoms are detected.	Integration
FR-14	The summary indicates whether the symptom, X-ray, and report results agree or differ.	Integration

Non-functional requirements (how it must perform)

ID	Requirement	Category
NFR-01	Summary page appears within about 10 seconds of submission.	Performance
NFR-02	Language is simple and understandable by non-medical users.	Usability
NFR-03	The site works in modern browsers on desktop and mobile screens.	Compatibility
NFR-04	Uploaded files and entered text are not stored permanently.	Privacy
NFR-05	All experiments are logged in MLflow with an agreed naming convention.	Reproducibility
NFR-06	Code is organised on GitHub with instructions on how to run it.	Maintainability
NFR-07	Model quality is reported with Accuracy, Precision, Recall, and F1 on a held-out test set.	Quality
NFR-08	Invalid input (wrong file type, empty text) produces a clear error message.	Reliability

Project constraints

ID	Constraint
C-01	Delivery within 4 weeks.
C-02	Only free, public datasets are used; no real patient data is collected.
C-03	Scope is limited to chest and respiratory conditions.
C-04	The X-ray model can distinguish only normal and pneumonia, based on the dataset used.
C-05	Free or student cloud tiers and free GPU notebooks only.
C-06	The product is a preliminary guide and must never present itself as a diagnosis.

5. Analyze and Prioritize

Each requirement is checked for feasibility, value, and dependencies, then ranked with MoSCoW.

Priority	Requirements	Reason and dependencies
Must have	FR-01, FR-03, FR-05, FR-06, FR-08, FR-09, FR-11, FR-12, NFR-04, NFR-05, NFR-07	The core of the idea: all three inputs, one summary, disclaimer, and tracked experiments. Datasets are available and tools are known.
Should have	FR-02, FR-04, FR-10, FR-13, FR-14, NFR-01, NFR-02, NFR-03, NFR-06, NFR-08	High value and feasible. FR-04 depends on FR-03; FR-14 depends on FR-03, FR-06, and FR-09.
Could have	FR-07 (heatmap)	Improves trust but depends on a finished X-ray model; added if Week 3 allows.
Won't have (this version)	Other specialties, user accounts, appointment booking, voice input, mobile app, additional X-ray findings	Out of scope for 4 weeks and not supported by the chosen data.

Key dependencies: the website needs all three models; the specialty mapping needs the symptom model; the consistency check needs all three results; deployment needs the final saved models.

6. Verify and Validate

Walkthrough of the page flow (wireframe description)

- **Page 1 – Symptoms:** a text box for symptoms; extracted symptoms are shown back to the user for confirmation.
- **Page 2 – X-ray:** image upload with preview.
- **Page 3 – Report:** text box or file upload for the report.
- **Page 4 – Summary:** symptoms, possible diseases, X-ray opinion and heatmap, report result and explained terms, suggested specialty, any urgent warning, download button.
- **All pages:** disclaimer banner on every page.

Wireframes of these four pages will be reviewed with the Project Owner and shown to a few potential users before development of the website starts (Week 3).

Validation checklist

- Every requirement is clear, testable, and has an ID.
- No requirement contradicts another or goes beyond the constraints.

- Every Must-have requirement maps to a component, an owner, and a week (Section 7).
- Scope and Won't-have items are understood by the Project Owner.

Sign-off

Name	Role	Signature	Date
	Project Owner (Instructor)		
	Team representative		

7. Manage Changes

Baseline

After sign-off, this document (version 1.0) is the baseline. Any later version records what changed, why, and who approved it.

What can change and what cannot

Not every change is equal. Changes are divided into three levels so that small technical decisions do not need approval, while changes that affect the Project Owner's expectations do.

Level	Examples	Process	Why
1. Frozen core	Project idea; the three inputs; chest/respiratory scope; one summary output; medical disclaimer; delivery time	Not changed. Only the Project Owner can approve an exception.	These define what was promised.
2. Change request	Cloud provider (for example, Azure to AWS); web framework (Streamlit to another); replacing a dataset; dropping a Must-have requirement	Written change request, impact on time and work, and Project Owner approval before the change.	The change affects accounts, deployment steps, cost, and the timeline.
3. Team decision	Using library Y instead of library X for the same function; model type; hyperparameters; code structure; extra experiments	Team decides and records the reason in MLflow or GitHub. No approval needed.	The outcome for the user is the same, and the Project Owner's requirements are unchanged.

Example: replacing one library with another that does the same job (Level 3) is free. Moving the whole deployment from Azure to AWS (Level 2) needs a change request, because it affects everything that was planned around Azure.

Change-control process (Level 2)

- **Step 1:** Submit a change request: what, why, and who asked.
- **Step 2:** Assess impact on scope, schedule, work, and other requirements.
- **Step 3:** Project Owner approves or rejects.
- **Step 4:** If approved, update this document to a new version and the traceability matrix; inform all team members.

Traceability matrix

Requirement	Component	Owner	Week	How it is verified
FR-01, FR-02	Symptoms page and Arabic dictionary	Members 5, 6	3	Sample phrases are extracted correctly
FR-03, FR-04	Symptoms model and specialty mapping	Member 2	2–3	Test-set metrics; mapping reviewed
FR-05, FR-06, FR-07	X-ray model and upload page	Members 3, 4, 6	2–4	Test-set metrics; error analysis; heatmap check
FR-08, FR-09, FR-10	Report model and term explanation	Member 5	2–3	Test-set metrics; term explanation review
FR-11, FR-12, FR-13, FR-14	Summary page and integration	Member 6 (with all)	3–4	End-to-end test cases
NFR-01 to NFR-08	Whole system	All members	1–4	MLflow log, GitHub, performance and error tests
Data for all models	Datasets and splits	Member 1	1	Data quality and plots report